

Challenge (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 4
- (C) 16
- (D) 32
- (E) 2

Q2. Convert $\sqrt[3]{27}$ to rational exponent form

- (A) $3^{1/3}$
- (B) $3^{1/2}$
- (C) $3^{2/3}$
- (D) 3^3
- (E) $3^{3/2}$

Q3. A hiker travels $\sqrt{36}$ miles in 2 hours. What is her average speed in miles per hour?

- (A) 6 mph
- (B) 4 mph
- (C) 8 mph
- (D) 18 mph
- (E) 12 mph

Q4. A wildlife refuge has $\sqrt{121}$ acres of land. How many acres are there?

- (A) 10 acres
- (B) 11 acres
- (C) 12 acres
- (D) 15 acres
- (E) 16 acres

Q5. Simplify $\sqrt[3]{-8}$

- (A) -2
- (B) 2
- (C) 4
- (D) 8
- (E) 16

Q6. A vehicle consumes $\sqrt{16}$ gallons of fuel per hour. How many gallons will it consume in 4 hours?

- (A) 8 gallons
- (B) 16 gallons
- (C) 32 gallons
- (D) 4 gallons
- (E) 2 gallons

Q7. A population of animals grows at a rate of $\sqrt{9}$

- (A) 1
- (B) 3
- (C) 9
- (D) 12
- (E) 15

Q8. Simplify $\sqrt{100}$

- (A) 10
- (B) 8
- (C) 12
- (D) 5
- (E) 20

Open Ended Questions (FRQ)

Q9. A park has a hiking trail that is $\sqrt{225}$ feet long. If a hiker walks $\frac{1}{3}$ of the trail, what distance does the hiker cover? Show your work and simplification.

Q10. A fuel tank can hold $\sqrt[3]{1000}$ gallons of fuel. Convert this to rational exponent form and calculate the actual volume of the tank. Show your work and calculations.

Chef's Tips

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- Chef's Tips: Ensure students understand the concept of radicals and their applications.
- Chef's Tips: Emphasize the importance of simplifying radicals and converting between radical and rational exponent forms.
- Chef's Tips: Encourage students to check their work and provide detailed solutions for free response questions.